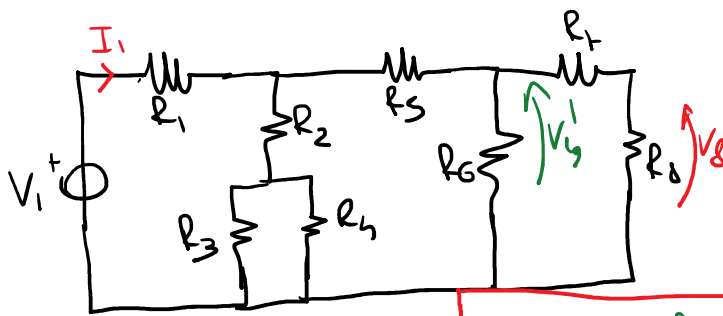


Esercizio 1

venerdì 18 marzo 2022 10:11



$$V_1 = 15V$$

$$R_1 = 10\Omega$$

$$R_2 = 25\Omega$$

$$R_3 = 50\Omega$$

$$R_4 = 20\Omega$$

$$R_5 = 24\Omega$$

$$R_6 = 20\Omega$$

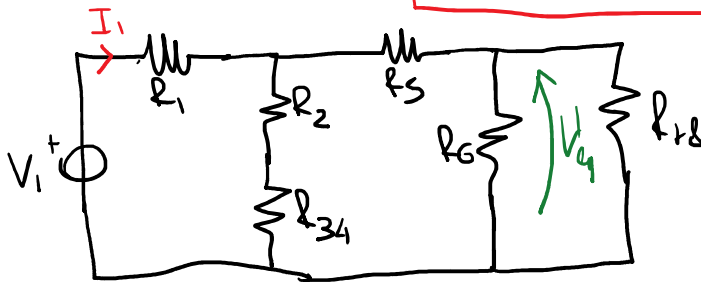
$$R_7 = 50\Omega$$

$$R_8 = 30\Omega$$

$$V_g = V_{eq}' \cdot \frac{R_8}{R_7 + R_8} = 2,47V$$

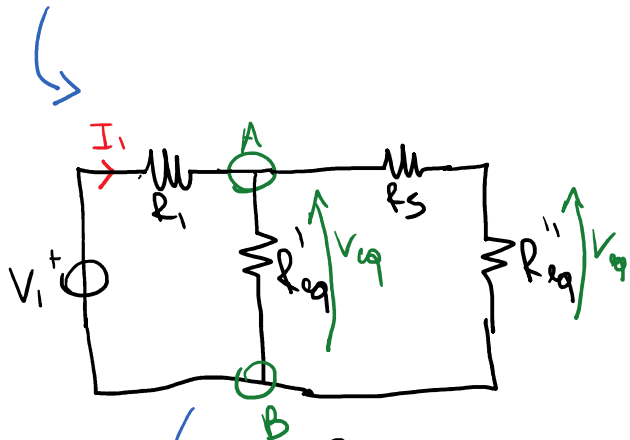
$$I_1 = ?$$

$$V_d = ?$$



$$R_{34} = R_3 // R_4 = 14,3\Omega$$

$$R_{78} = 80\Omega$$

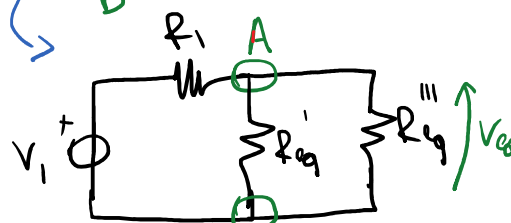


$$R_{eq}' = R_2 + R_{34} = 39,3\Omega$$

$$R_{eq}'' = R_6 // R_{78} = 16\Omega$$

$$V_{eq}' = V_{eq} \cdot \frac{R_{eq}''}{R_5 + R_{eq}''} = 6,6V$$

$$R_{eq}''' = R_5 + R_{eq}'' = 40\Omega$$



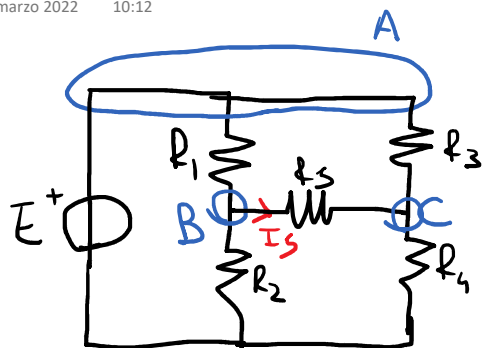
$$R_{eq} = R_{eq}' // R_{eq}''' = 19,8\Omega$$

$$I_1 = \frac{V_1}{R_1 + R_{eq}} \approx 0,5A$$

$$V_{eq} = I_1 \cdot R_{eq} = 9,9V$$

Esercizio 2

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$$R_1 = 10 \Omega$$

$$R_2 = 40 \Omega$$

$$R_3 = 20 \Omega$$

$$R_4 = 5 \Omega$$

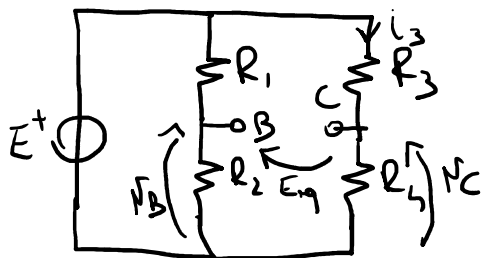
$$R_5 = 12 \Omega$$

$$E = 10 \text{ V}$$

$$I_S = ?$$

Thauevin Gra i nodi B e C

$$E_{eq}$$



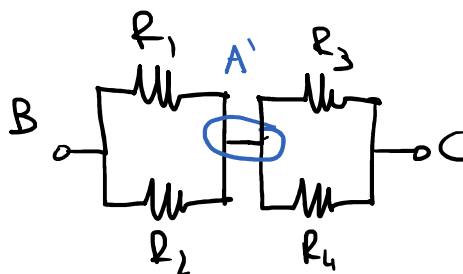
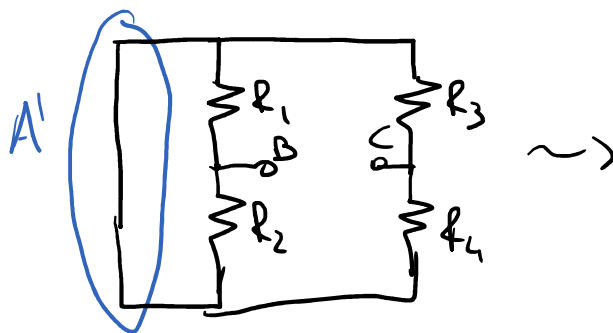
$$E_{eq} = V_{BC} = V_B - V_C$$

$$V_B = E \cdot \frac{R_2}{R_1 + R_2}$$

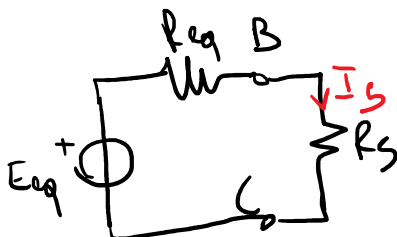
$$I_3 = \frac{E}{R_3 + R_4} \rightarrow V_C = R_4 \cdot I_3$$

$$E_{eq} = E \cdot \frac{R_2}{R_1 + R_2} - E \cdot \frac{R_4}{R_3 + R_4} = 6 \text{ V}$$

$$R_{eq}$$



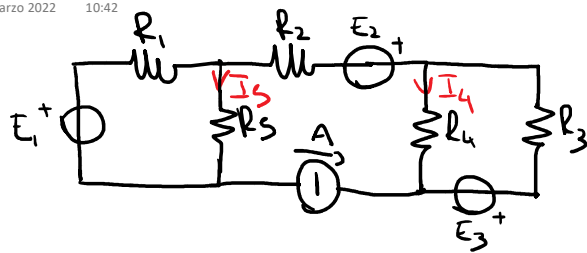
$$R_{eq} = R_1 // R_2 + R_3 // R_4 = 12 \Omega$$



$$I_S = \frac{E_{eq}}{R_{eq} + R_5} = \frac{6}{24} = 0,25 \text{ A}$$

Esercizio 3

venerdì 18 marzo 2022 10:42



$$E_1 = 50 \text{ V}$$

$$R_1 = R_5 = 5 \Omega$$

$$E_2 = 30 \text{ V}$$

$$R_2 = 35 \Omega$$

$$E_3 = -20 \text{ V}$$

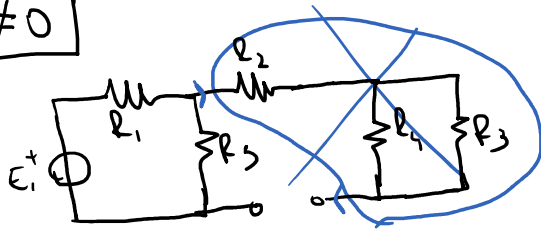
$$R_3 = R_4 = 10 \Omega$$

$$A = 10 \text{ A}$$

Calcolare la potenza assorbita da R_4 e R_5

SOVRAPPOSIZIONE DEGLI EFFETTI

$$E_1 \neq 0$$



$$P_4 = R_4 I_4^2$$

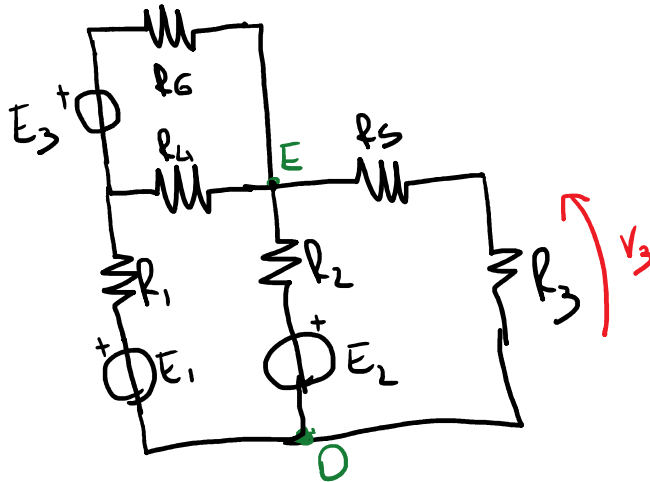
$$P_5 = R_5 I_5^2$$

$$I_4' = 0$$

$$I_5' = \frac{E_1}{R_1 + R_5}$$

Esercizio 4

venerdì 18 marzo 2022 11:14



Thevenin E-O

$$E_1 = 40 \text{ V}$$

$$E_2 = 25 \text{ V}$$

$$E_3 = 100 \text{ V}$$

$$R_1 = 4 \Omega$$

$$R_2 = 15 \Omega$$

$$R_3 = 4 \Omega$$

$$R_4 = 15 \Omega$$

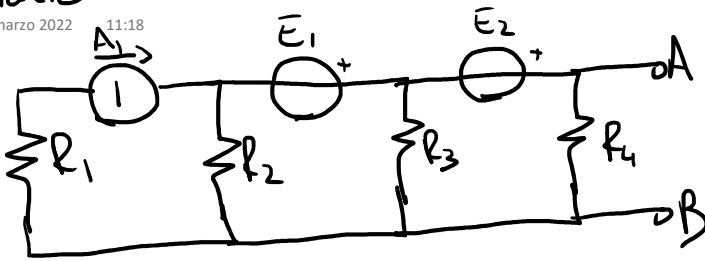
$$R_5 = 4 \Omega$$

$$R_6 = 10 \Omega$$

$$V_3 = ?$$

Esercizio 5

venerdì 18 marzo 2022 11:18



$$I_1 = 2A$$

$$R_1 = 5\Omega$$

$$E_1 = 10V$$

$$R_2 = R_3 = 10\Omega$$

$$E_2 = -20V$$

$$R_4 = 40\Omega$$

Norton tra A e B